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GEN AI ARCHITECT JOURNEY · PHASE-003

PyAutoGUI Automation

Desktop RPA - driving the mouse, keyboard, and screen

Track	PyAutoGUI - desktop automation
Contents	Four fundamentals scripts and two complete bots
Level	Beginner to Intermediate
Platform	Windows 10 or 11 · Python 3.10+
Folder	Phase-003-Robotic-Process-Automation-(RPA)/PyAutoGUI Automation

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1. What PyAutoGUI does

PyAutoGUI simulates physical mouse and keyboard actions.

Because it drives the real input devices, it can work with browsers and native Windows applications that expose no automation API at all. That generality is also its weakness: input goes to whichever window currently has focus, so the desktop has to be left alone for the whole run.

2. The fundamentals scripts

Script	What it demonstrates
basic_pyautogui.py	Imports PyAutoGUI, enables the upper-left-corner fail-safe, and adds a one-second pause after each action
pyautogui_mouse_operation.py	Move, click, double-click, right-click, and middle-click at (100, 100)
pyautogui_keyboard_operation.py	Types into the active application, then select, copy, Enter, and paste
pyautogui_screenshot_operation.py	Captures the complete screen to screenshot.png

3. Setup

```
POWERSHELL
cd "PyAutoGUI Automation"
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install pyautogui pyperclip pillow
```

Pillow is required by PyAutoGUI's screenshot feature.

4. Running the examples

```
POWERSHELL
python basic_pyautogui.py
python pyautogui_mouse_operation.py
python pyautogui_keyboard_operation.py
python pyautogui_screenshot_operation.py
```

Run one example at a time. Before the keyboard example, open Notepad or another safe text editor - the script gives you five seconds to focus it. The mouse example operates near the upper-left of the screen immediately, so use it only when clicks at those coordinates are harmless.

5. Behaviour worth knowing

- Coordinates are **absolute screen pixels** and behave differently under another resolution, monitor layout, or display scale.
- Keyboard input goes to whichever window has focus - not to the script.
- **pyautogui_screenshot_operation.py** overwrites **screenshot.png** when run again from this directory.
- The standalone examples do not all configure **FAILSAFE**; use **Ctrl + C** from the terminal if you need to stop one.



The fail-safe

In the two complete bots, PyAutoGUI's fail-safe is enabled: move the pointer quickly to the upper-left corner of the primary display and the automation aborts. Learn this before running anything that types.

6. The two complete bots

Project	What it automates
Assignment-1-PyAutoGUI_Weather_Data	AccuWeather -> parsed readings -> dated Notepad file
Assignment-2-PyAutoGUI_Bot_Stock_Price	Google Finance -> formatted Excel report -> screenshot

Each carries its own README and PDF with full setup, output, and troubleshooting detail.

Next

Continue to the Bengaluru Weather Bot, the first complete desktop automation in this track.

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